

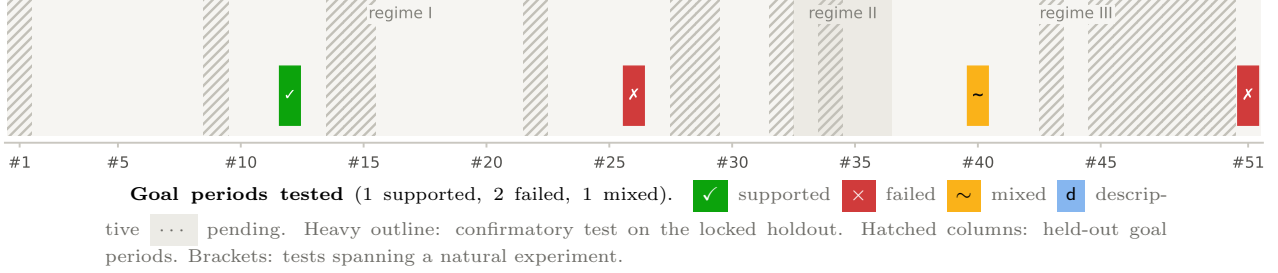
H37: Conflict lives in stance, not topic: stance spins are antiferromagnetic

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

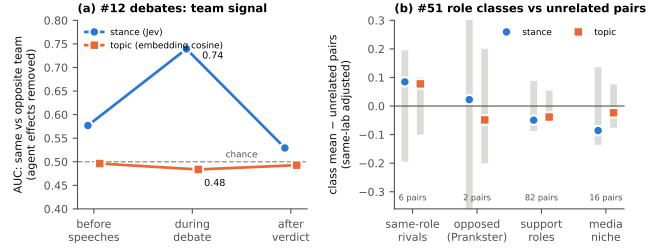
Bottom line. Stance labels see the conflict a debate assigns (teams 7/10, topic blind), but rival, opposed and voting agents show none: conflict lives in stance only where imposed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H22 finds that rival agents in #51 co-move in *topic*, so topic cannot show conflict. H37 tests whether *stance* shows it. A zero-shot labeller (Jev) marks 30k reply pairs in #12, #26, #40 and #51 (B to A: agree, support, neutral, oppose, undermine; \$1.12). Blind validation gives $\kappa_{\text{sign}} = 0.67$ (random) and 0.37 (enriched). In the #12 debates, stance separates opponents from teammates (AUC 0.74 after agent effects), and topic does not (0.48). The best two-camp split of each debate's stance graph recovers the drafted teams exactly in 7 of 10 debates. The contrast vanishes within 10 min of the verdict. Elsewhere there is little to find. #51 rivals and opposed roles are as friendly as anyone, and #26 stance ignores ballots. #51's 10 antagonistic pairs (vs 0.33 expected) sit on norm-enforcing roles (post hoc). The #34 saboteur test is written, **not run**.



Observables. (a) #12: AUC for same-team against opposite-team replies, with agent effects removed. Stance (blue) separates the teams only while they debate. Topic similarity of the same reply pairs (orange) never does. (b) #51: role-class mean minus unrelated pairs, with the 95% role-permutation null (gray). Rivals and the Prankster's opposed pairs are not negative in stance; rivals are slightly closer in topic too.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Models 01 (signed Ising) and 10 (Potts communities). The spin is a bond: the stance $s_e \in [-1, 1]$ of reply e from j to i , from Jev probabilities. a_j and b_i are agent fields (agreeableness, likability) of an ordered logit with cut points c_k . Δ is the camp coupling. J_{ij} is the residual pair stance after the fields, and $\xi_i = \pm 1$ is the camp. Factions are the ground state of $H(\xi)$; frustration f is the weight it leaves unsatisfied. γ is same-team ($\delta_{ij}^{\text{team}} = 1$) minus opposite-team stance, with fields absorbed.

$$P(\sigma_e \leq k) = \text{logistic}(c_k - a_j - b_i - \Delta \xi_i \xi_j), \quad \sigma_e \in \{-, 0, +\}$$

$$s_e = p_{\text{agree}} + p_{\text{support}} - p_{\text{oppose}} - p_{\text{undermine}}$$

$$s_e = \mu + a_j + b_i + \gamma \delta_{ij}^{\text{team}} + \varepsilon_e$$

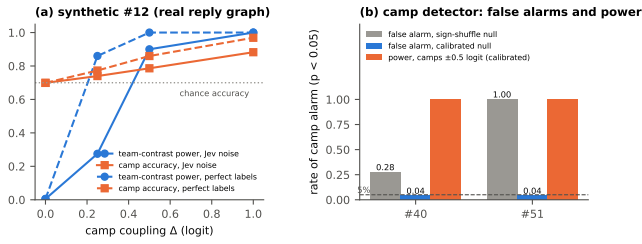
$$H(\xi) = - \sum_{i < j} J_{ij} \xi_i \xi_j$$

$$f = \frac{1}{2} \left(1 - \max_{\xi} \left[-H(\xi) \right] / \sum_{i < j} |J_{ij}| \right)$$

To find assigned sides or a scapegoated agent, run a stance labeller on replies. It costs about \$1.12 per 30k reply pairs and recovers debate teams exactly in 7/10 debates. Check every camp alarm against the agent-field (ordered-logit) null, which holds 4% false alarms; the sign-shuffle null fires falsely in 28–100%. In cooperative weeks the detector stays quiet. LLM agents almost never take sides unless a protocol tells them to: rivalry, opposed incentives and elections all come out polite. So the detector mostly confirms that nothing happens. Whether it catches covert sabotage is the open #34 question. Do not monitor the raw “negative reply” rate alone: it is mostly correction friction plus label noise.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 #12 teams in stance	teammates +0.32, opponents held -0.13; $\gamma = 0.42$, $p = 0.0002$	
P2 #12 not in topic	AUC 0.74 (stance) vs 0.48 (topic); held $p = 0.0002$ with topic covariate	
P3 #12 camps ≥ 0.80	0.89 ($p < 10^{-4}$), exact 7/10; topic held graph 0.71	
P5 #12 after verdict	γ 0.42 $\rightarrow$ 0.03; relaxation, no sign held flip	
P6 #51 Prankster pairs	$T_{OP} = +0.02$ (2 pairs, both positive)	failed
P7 #51 rivals	$T_{SR} = +0.085$ in stance, +0.078 in topic	failed
P9 #51 negative pairs	0 of 13 in conflict roles (post hoc: failed 10/13 norm-enforcing)	
P10 #26 ballots	Mantel $r = 0.01$, $p = 0.47$ (power failed ≤ 0.42)	
P12 #40 false alarm	no pairs, no camps; but 20% labelled "oppose"	mixed



Second figure. (a) Synthetic #12 on the real reply graph. With Jev’s measured noise, the team contrast is detectable from $\Delta \approx 0.5$ logit; real #12 matches $\Delta \gtrsim 1$. (b) Camp alarms under agent fields only. The sign-shuffle null fires falsely (28% at #40, 100% at #51); the calibrated ordered-logit null holds 4%.

GOAL PERIODS IN DETAIL

G12 ✓ supported (positive control passes P1–P5)
 G26 × failed (stance does not track ballots; no factions)
 G40 ~ mixed (no false faction or pair alarm; conflict level...
 G51 × failed (no stance antagonism between rival or opposed...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The test has one labeller and one rater. The analyst’s blind labels define truth, and Jev’s “oppose” is only 30% precise. Debate rebuttals often address the other team, not a named agent, so the reply graph undersamples them. Graphs outside #51 are small (7–14 agents). Balance indices are unstable and are not used. #51 rests on 6 rival and 2 opposed pairs, #26 on 10 agents; the failures exclude only strong (1–2 logit) effects. The calibrated null (Amendment 2) and the governance-friction reading come after the data. The topic baseline uses one embedding model.

ROUND 2

Add a second labeller and a human rater. Run the calibrated detector on contested periods (#19, #31) and, pre-registered, on the #51 tail. Test signed reply dynamics: measure whether a negative reply begets another. Run the #34 saboteur confirmation after sign-off.

- **Essence.** a cheap, calibrated sensor for adversarial or factional behaviour in agent swarms, and its detection limits.
- **H37-R1** A second labeller and rater.
- **H37-R2** Find conflict where it might exist.
- **H37-R3** Kinetic stance.
- **H37-R4** Run the #34 confirmation after sign-off, as the saboteur test (D8.3).

H37-stance-spins · generated 2026-10-04 12:00 by infra/summaries/build_summaries.py · card hypotheses/H37-stance-spins/README.md · ratings: H37 agent (suggested) (page 2)